MAJOR PROJECT · SEMESTER 7

Strategic Application of AI in CAD

Plain English in, checked machine joints out. Type a sentence, get a designed, checked and drawn machine joint.

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THE PROBLEM

Designing a joint by hand is slow, and the slow part is not the drawing

For one knuckle joint a student computes the rod from tension, sizes the pin for shear and again for bending, takes the larger, derives seven proportions from the rod, then checks nine separate failure modes against allowable stresses.

Change the load and all of it is done again from the start.

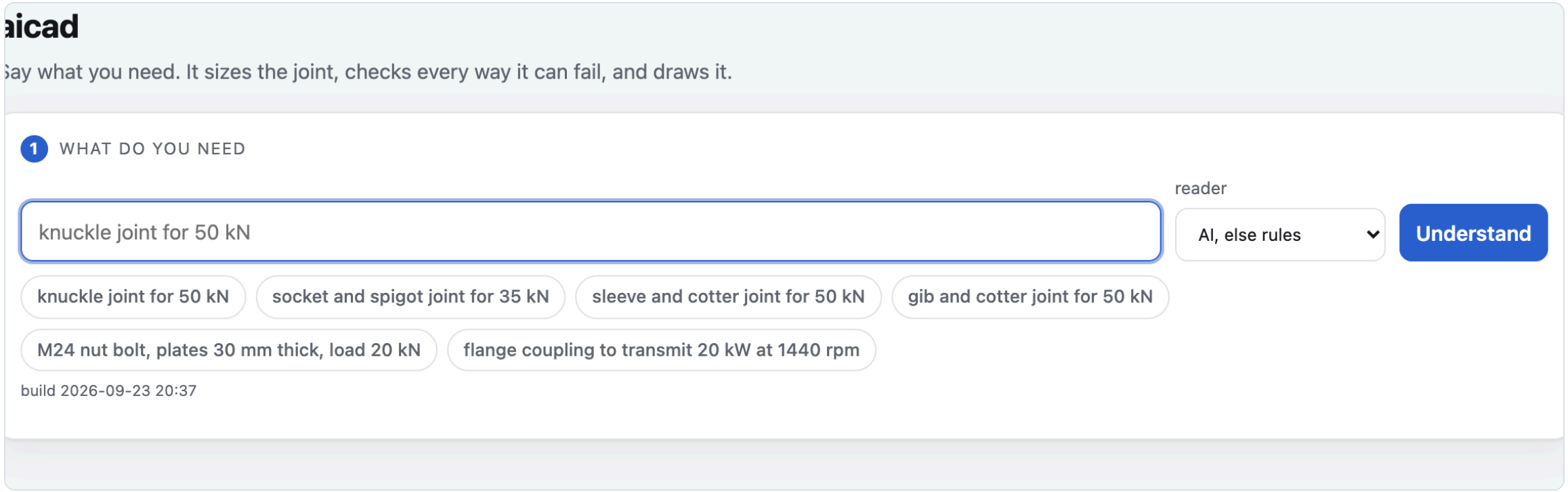
> **The aim:** keep every formula exactly as the textbook has it, and remove only the repetition.

DONE BY HAND, PER JOINT
Rod diameter from tension
Pin from shear, then from bending
Seven proportions from the rod
Nine failure checks
Re-size anything that fails
Draw four views and dimension them

WHAT WAS BUILT

A sentence goes in. A checked design comes out.

6	44	290	10s
JOINTS	FAILURE CHECKS	TESTS	PER DESIGN



The live site. Knuckle, socket and spigot cotter, sleeve and cotter, gib and cotter, hex bolt and nut, and a protected type rigid flange coupling.

THE DESIGN DECISION

The AI reads the request. It does not choose a single dimension.

Turning "50 kN" into 50,000 newtons and knowing that "socket and spigot" names a cotter joint is a language problem. A language model is good at that.

Choosing a pin diameter is an engineering problem with a correct answer. That stays with deterministic code, traceable to a textbook page.

> **If asked why the AI does not generate the whole model:** because then no number could be checked against anything. Here every dimension has a formula and every formula has a source.

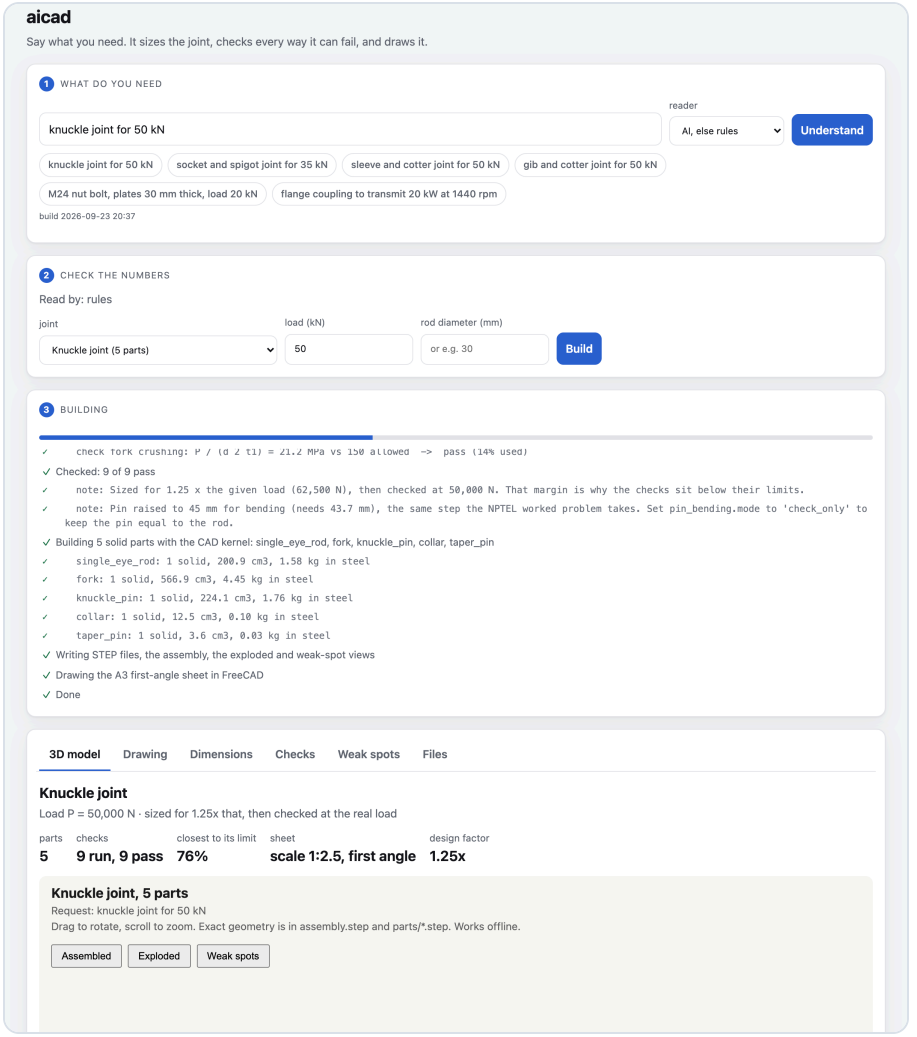
STAGE	WHO DOES IT
Read the sentence	**AI**
Size every dimension	formula
Check every failure	formula
Build the solids	CAD kernel
Draw the sheet	FreeCAD

WORKED EXAMPLE

Knuckle joint, 50 kN, shown as the software computes it

FORMULA RESULT	MM	OUTCOME
`rod in tension`	32.57	rod becomes 35
`pin in double shear`	25.75	not governing
`pin in bending`	43.72	**pin becomes 45**

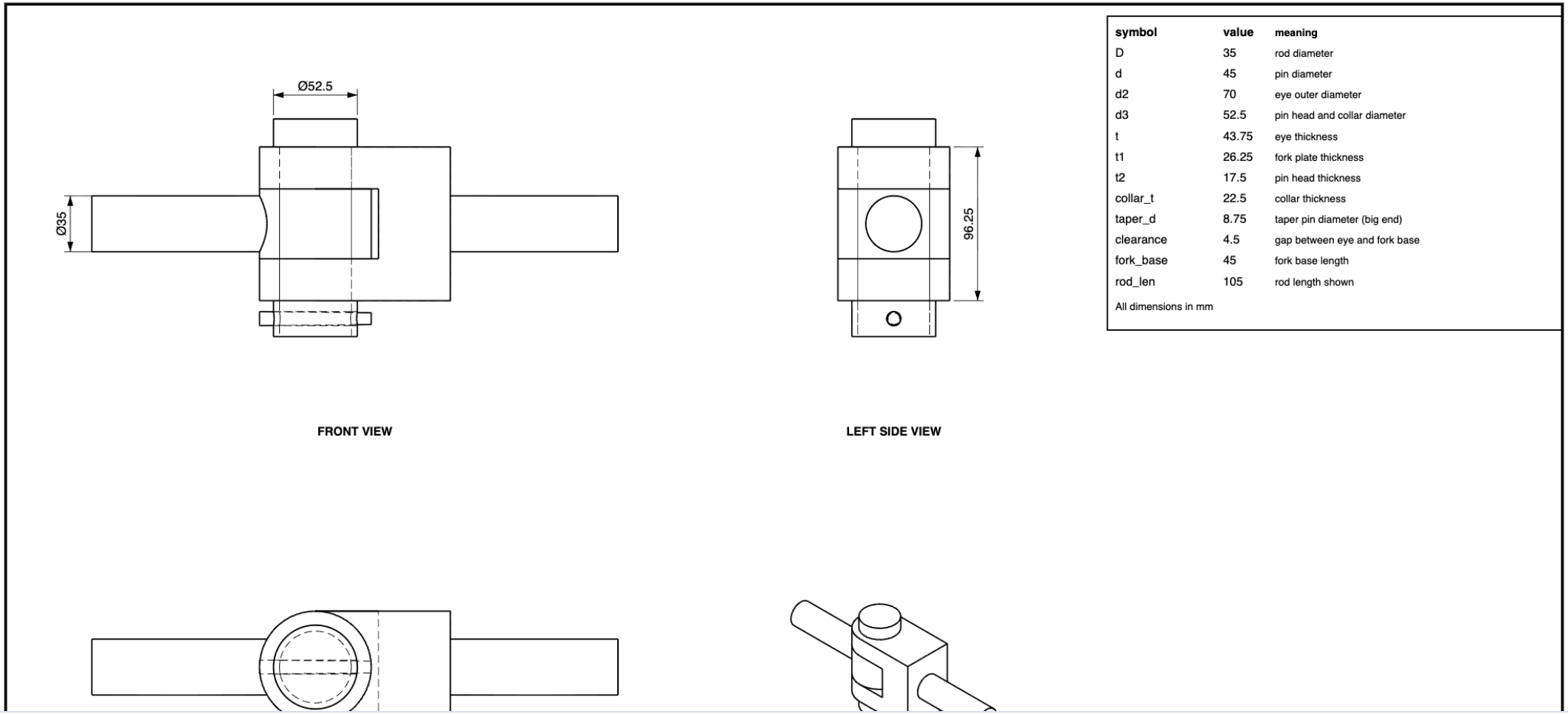
> Sized for 1.25 times the load, then reported at the real load. Busiest check sits at **76 percent** of its allowable, not at 100, so there is margin in hand.



Every run prints its working, so the output can be checked rather than trusted.

OUTPUT

A first angle A3 sheet, dimensioned, with a title block



symbol	value	meaning
D	35	rod diameter
d	45	pin diameter
d2	70	eye outer diameter
d3	52.5	pin head and collar diameter
t	43.75	eye thickness
t1	26.25	fork plate thickness
t2	17.5	pin head thickness
collar_t	22.5	collar thickness
taper_d	8.75	taper pin diameter (big end)
clearance	4.5	gap between eye and fork base
fork_base	45	fork base length
rod_len	105	rod length shown

All dimensions in mm

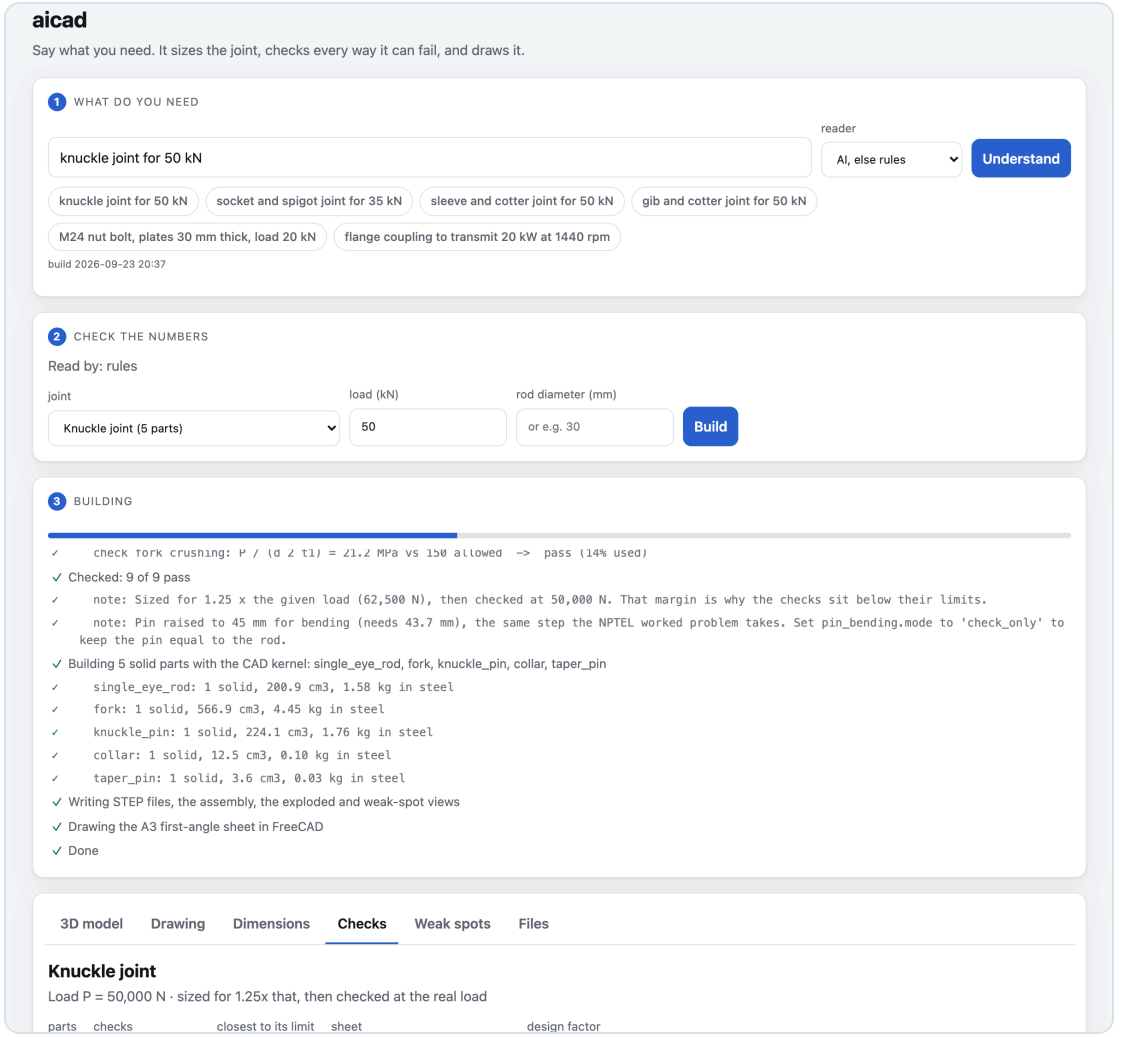
Produced by FreeCAD driven without its interface. The same views are also written as a DXF for AutoCAD, and every part as a STEP file that opens in NX.

VERIFICATION

Checked against published books, and the books lost three times

Sources: NPTEL IIT Kharagpur Design of Machine Elements I, Unwin 1899, and Leutwiler 1917. The last two are out of copyright.

- **The knuckle lesson** raises the pin for bending, then checks the eye with the old pin. Its own answer fails its own check, 80 MPa against 65 allowed.
- **The cotter lesson** omits crushing between cotter and spigot. At its answer that reads 167 MPa against 110 allowed. A second book lists it.
- **A printed constant** is off by a factor of ten, disproved by the same page's own worked example.

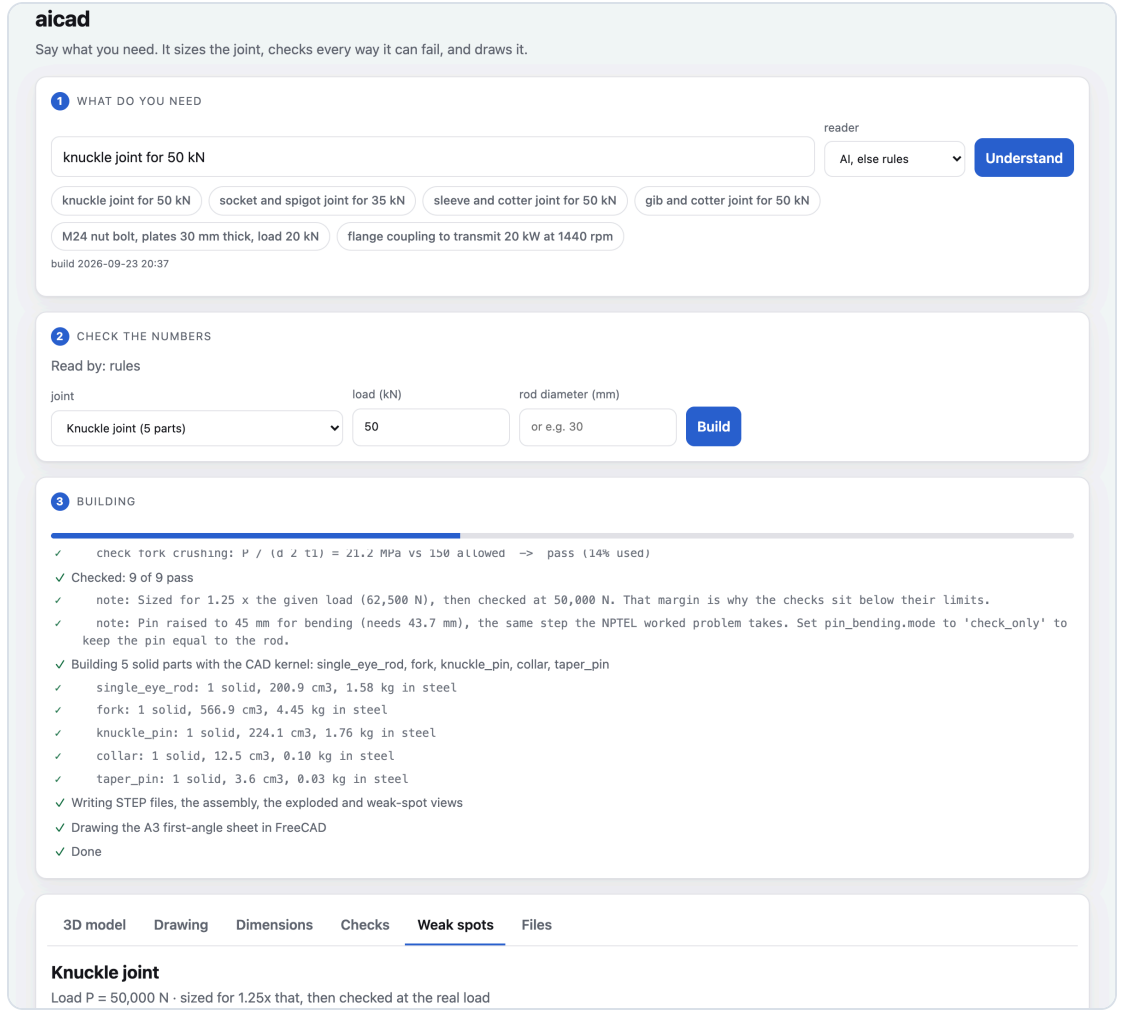


Each check shows its substituted arithmetic, which is what made the disagreements visible.

HONEST LIMITS

What this does not do

- The stress analysis is one linear static run on one part. It is a second opinion on the formula, not a replacement for proper analysis.
- Proportions for two of the six joints have no book behind them, and the software says so on every run.
- Fatigue, stress concentration, tolerances and fits are out of scope. Sizing is static.
- The published site reads with pattern rules, because the server holds no API key.



Parts are coloured by how close each is to its limit, so the weakest member is visible rather than buried in a table.

IN ONE LINE

Every dimension it produces can be traced to a formula, and every formula to a page in a book.

That is the difference between a tool that designs and one that only draws. The AI reads the request; the engineering stays checkable.

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